UpSkill Sprint Consulting

QUICK REFERENCE · ASQ CSSBB · CQE · CMBB

# DMAIC Formula Encyclopedia

## Every formula, organised by phase

111 formula sets from Pre-DMAIC through Control: each with the equation, when to use it and the exam trap to avoid. Colour-coded by phase so you can find the right formula in seconds.

| 111 | 87 | 6 | 3 |
|---|---|---|---|
| FORMULA SETS | HIGH-YIELD | COLOUR-CODED SECTIONS | ASQ EXAMS COVERED |

| 9 | 4 | 19 | 52 | 16 | 11 |
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| Pre-DMAIC | Define | Measure | Analyze | Improve | Control |
| P01–P09 | D01–D04 | M01–M19 | A01–A52 | I01–I16 | C01–C11 |

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111 formula sets across six colour-coded sections. Each section starts on a new page; the coloured tab on the page edge shows where you are.


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**READING A CARD**

M09 Formula ID: the phase letter plus a number, matching the online lesson.

★ High-yield Tested often. Learn these first.

CSSBB CQE CMBB Exams whose Body of Knowledge covers the formula.

**EACH CARD GIVES YOU**

**Equation** Shaded panel, in the phase colour.

**When to use** The situation the formula answers.

**Exam trap** The mistake that costs marks.

# Formula index

Every formula set by ID. Click any row to jump straight to that formula; every page footer links back here.


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PRE

BEFORE THE PROJECT · P01–P09

# Pre-DMAIC

Enterprise deployment, finance, project selection and planning.

9 FORMULA SETS

★ 7 high-yield



## P01 Revenue growth, market share and profit margins

BUSINESS MEASURES · CSSBB · CQE · CMBB · ★ High-yield

$$\text{Revenue Growth \%} = \frac{R_t - R_{t-1}}{R_{t-1}} \times 100\%$$

$$\text{Market Share \%} = \frac{\text{Organization Sales}}{\text{Total Market Sales}} \times 100\%$$

$$\text{Gross Margin \%} = \frac{\text{Revenue} - \text{COGS}}{\text{Revenue}} \times 100\%$$

$$\text{Operating Margin \%} = \frac{\text{Operating Income}}{\text{Revenue}} \times 100\%$$

$$\text{Net Margin \%} = \frac{\text{Net Income}}{\text{Revenue}} \times 100\%$$

**WHEN TO USE**

Translate strategic objectives into measurable financial outcomes.

**EXAM TRAP**

Margin is based on revenue, while markup is based on cost. They are not interchangeable.

## P02 Contribution margin, profit and break-even analysis

FINANCE · CSSBB · CQE · CMBB · ★ High-yield

$$CM_u = P - V$$

$$CMR = \frac{P - V}{P}$$

$$\Pi = (P - V)Q - F$$

$$Q_{BE} = \frac{F}{P - V}$$

$$S_{BE} = \frac{F}{CMR}$$

$$Q_{\text{target}} = \frac{F + \Pi_{\text{target}}}{P - V}$$

**WHEN TO USE**

Evaluate project economics, production alternatives and target-profit requirements.

**EXAM TRAP**

Use variable cost per unit, not total variable cost, in the unit break-even denominator.

## P03 Present value, future value and annuities

FINANCE · CSSBB · CQE · CMBB · ★ High-yield

$$FV_n = PV_0(1 + i)^n$$

$$PV_0 = \frac{FV_n}{(1 + i)^n}$$

$$PV_{\text{annuity}} = A\left[\frac{1 - (1 + i)^{-n}}{i}\right]$$

$$FV_{\text{annuity}} = A\left[\frac{(1 + i)^n - 1}{i}\right]$$

$$PV_{\text{perpetuity}} = \frac{A}{i}$$

**WHEN TO USE**

Convert cash flows occurring at different times to a common economic basis.

**EXAM TRAP**

The interest rate and number of periods must use matching time units.

## P04 NPV, IRR, ROI, benefit–cost ratio and payback

FINANCE · CSSBB · CQE · CMBB · ★ High-yield

$$NPV = \sum_{t=0}^{n} \frac{CF_t}{(1+i)^t}$$

$$0 = \sum_{t=0}^{n} \frac{CF_t}{(1+IRR)^t}$$

$$ROI = \frac{\text{Benefit} - \text{Cost}}{\text{Cost}} \times 100\%$$

$$BCR = \frac{\sum PV(\text{Benefits})}{\sum PV(\text{Costs})}$$

$$\text{Simple Payback} = \frac{\text{Initial Investment}}{\text{Annual Net Cash Inflow}}$$

**WHEN TO USE**

Compare improvement projects and capital alternatives.

**EXAM TRAP**

A benefit–cost ratio greater than one is economically favorable, but it does not prove it is the best available project.

## P05 Cost of quality and activity-based costing

COST OF QUALITY · CSSBB · CQE · CMBB · ★ High-yield

$$COQ = C_P + C_A + C_{IF} + C_{EF}$$

$$COPQ = C_{IF} + C_{EF}$$

$$\text{Activity Rate} = \frac{\text{Activity Cost Pool}}{\text{Total Cost-Driver Quantity}}$$

$$\text{Allocated Activity Cost} = \text{Activity Rate} \times \text{Driver Quantity Consumed}$$

$$\text{Unit Cost} = \frac{\text{Direct Cost} + \text{Allocated Overhead}}{\text{Units Produced}}$$

**WHEN TO USE**

Quantify prevention, appraisal and failure costs and assign overhead using actual process drivers.

**EXAM TRAP**

COPQ commonly means failure-related poor-quality cost; organizational definitions may also include selected appraisal or opportunity costs.

## P06 Earned value management

PROJECT MANAGEMENT | CSSBB | CMBB | ★ High-yield

$$CV = EV - AC$$

$$SV = EV - PV$$

$$CPI = \frac{EV}{AC}$$

$$SPI = \frac{EV}{PV}$$

$$EAC_1 = \frac{BAC}{CPI}$$

$$EAC_2 = AC + (BAC - EV)$$

$$EAC_3 = AC + \frac{BAC - EV}{CPI \times SPI}$$

$$ETC = EAC - AC$$

$$VAC = BAC - EAC$$

$$TCPI_{BAC} = \frac{BAC - EV}{BAC - AC}$$

**WHEN TO USE**

Monitor whether a project is obtaining planned value at the planned cost and schedule.

**EXAM TRAP**

Positive schedule variance is favorable. A CPI or SPI less than one is unfavorable.

## P07 PERT, CPM and activity slack

PROJECT MANAGEMENT | CSSBB | CQE | CMBB | ★ High-yield

$$t_e = \frac{a + 4m + b}{6}$$

$$\sigma_t^2 = \left(\frac{b - a}{6}\right)^2$$

$$\mu_{\text{path}} = \sum_i t_{e,i}$$

$$\sigma_{\text{path}}^2 = \sum_i \sigma_{t,i}^2$$

$$Z = \frac{T - \mu_{\text{critical path}}}{\sigma_{\text{critical path}}}$$

$$\text{Total Slack} = LS - ES = LF - EF$$

$$\text{Free Slack} = ES_{\text{successor}} - EF_{\text{activity}}$$

**WHEN TO USE**

Estimate uncertain activity durations and identify the critical path.

**EXAM TRAP**

The project variance is the sum of variances along the critical path, not the square of the sum of standard deviations.

## P08 Training ROI and benefit–cost evaluation

TRAINING EVALUATION | CSSBB | CMBB

$$\text{Net Program Benefit} = \text{Program Benefits} - \text{Program Costs}$$

$$ROI_{\text{training}} = \frac{\text{Program Benefits} - \text{Program Costs}}{\text{Program Costs}} \times 100\%$$

$$BCR_{\text{training}} = \frac{\text{Program Benefits}}{\text{Program Costs}}$$

**WHEN TO USE**

Evaluate whether knowledge and behavior changes justify the total training investment.

**EXAM TRAP**

Do not attribute all post-training improvement to training without isolating other causes.

## P09 Expected monetary value and weighted project scoring

RISK AND PORTFOLIO | CSSBB | CQE | CMBB

$$EMV = \sum_{i=1}^{k} p_i V_i$$

$$\text{Expected Loss} = P(\text{event}) \times \text{Loss if Event Occurs}$$

$$S_j = \sum_{i=1}^{m} w_i r_{ij}, \qquad \sum_i w_i = 1$$

$$\text{Risk-Adjusted Value} = \text{Expected Benefit} - \text{Expected Risk Cost}$$

**WHEN TO USE**

Prioritize projects using probability-weighted financial or strategic consequences.

**EXAM TRAP**

Weighted scoring is only meaningful when scales are aligned and the weights are established before reviewing project scores.

D

DMAIC PHASE 1 · D01–D04

# Define

Customer, business case, risk and project-definition metrics.

4 FORMULA SETS

★ 2 high-yield

## D01 Net promoter, retention and churn

CUSTOMER METRICS · CSSBB · CQE · CMBB

$$NPS = \%\text{Promoters} - \%\text{Detractors}$$

$$\text{Retention Rate} = \frac{E - N}{S} \times 100\%$$

$$\text{Churn Rate} = \frac{\text{Customers Lost During Period}}{\text{Customers at Start of Period}} \times 100\%$$

**WHEN TO USE**

Summarize customer loyalty and customer-base stability.

**EXAM TRAP**

In the retention equation, E is ending customers, N is newly acquired customers and S is starting customers.

## D02 Weighted decision matrix and QFD relationship score

DECISION ANALYSIS · CSSBB · CQE · CMBB · ★ High-yield

$$S_j = \sum_{i=1}^{m} w_i r_{ij}$$

$$W_j^{QFD} = \sum_{i=1}^{m} I_i R_{ij}$$

$$W_{j,\text{normalized}} = \frac{W_j}{\sum_j W_j} \times 100\%$$

**WHEN TO USE**

Rank project alternatives, technical requirements or solution concepts.

**EXAM TRAP**

Do not multiply ordinal ratings as though they were precise ratio-scale measurements without acknowledging the approximation.

## D03 Pareto percentage and cumulative percentage

PRIORITIZATION · CSSBB · CQE · CMBB

$$\%\text{Category}_i = \frac{x_i}{\sum_j x_j} \times 100\%$$

$$\%\text{Cumulative}_k = \frac{\sum_{i=1}^{k} x_{(i)}}{\sum_j x_j} \times 100\%$$

**WHEN TO USE**

Identify the largest contributors after arranging categories in descending order.

**EXAM TRAP**

The 80/20 relationship is a heuristic, not a mathematical requirement.

## D04 FMEA risk calculations

RISK ANALYSIS · CSSBB · CQE · CMBB · ★ High-yield

$$RPN = S \times O \times D$$

$$\text{Criticality} = \beta \lambda_p t$$

$$C_m = \beta \alpha \lambda_p t$$

$$\text{Risk Exposure} = P(\text{failure}) \times \text{Consequence}$$

**WHEN TO USE**

Prioritize potential failure modes and risk-reduction actions.

**EXAM TRAP**

Modern AIAG/VDA action priority does not rely solely on RPN. Equal RPN values can represent very different risk profiles.

DMAIC PHASE 2 · M01–M19

# M Measure

Data, measurement systems, descriptive statistics and baseline performance.

19 FORMULA SETS

★ 17 high-yield

## M01 Arithmetic, weighted, geometric and harmonic means

DESCRIPTIVE STATISTICS · CSSBB · CQE · CMBB · ★ High-yield

$$\bar{x} = \frac{1}{n}\sum_{i=1}^{n} x_i$$

$$\bar{x}_w = \frac{\sum_i w_i x_i}{\sum_i w_i}$$

$$G = \left(\prod_{i=1}^{n} x_i\right)^{1/n}$$

$$H = \frac{n}{\sum_{i=1}^{n} 1/x_i}$$

**WHEN TO USE**

Arithmetic mean for additive data, geometric mean for multiplicative growth and harmonic mean for averaged rates.

**EXAM TRAP**

Do not average rates with the arithmetic mean when each rate applies to an equal quantity rather than an equal time.

## M02 Median, range, quartiles and IQR fences

DESCRIPTIVE STATISTICS · CSSBB · CQE · CMBB · ★ High-yield

$$R = x_{\max} - x_{\min}$$

$$IQR = Q_3 - Q_1$$

$$LF = Q_1 - 1.5(IQR)$$

$$UF = Q_3 + 1.5(IQR)$$

$$\text{Extreme Lower Fence} = Q_1 - 3(IQR)$$

$$\text{Extreme Upper Fence} = Q_3 + 3(IQR)$$

**WHEN TO USE**

Summarize spread and screen for potential outliers without assuming normality.

**EXAM TRAP**

Values outside the fences are candidates for investigation, not automatic data-entry errors.

## M03 Population and sample variance, standard deviation and CV

DESCRIPTIVE STATISTICS · CSSBB · CQE · CMBB · ★ High-yield

$$\sigma^2 = \frac{\sum_{i=1}^{N}(x_i - \mu)^2}{N}$$

$$s^2 = \frac{\sum_{i=1}^{n}(x_i - \bar{x})^2}{n-1}$$

$$s = \sqrt{s^2}$$

$$CV = \frac{s}{|\bar{x}|} \times 100\%$$

$$s_p^2 = \frac{(n_1-1)s_1^2 + (n_2-1)s_2^2}{n_1 + n_2 - 2}$$

**WHEN TO USE**

Quantify variation and pool two variance estimates when equal variance is justified.

**EXAM TRAP**

The sample variance uses n−1. CV is unstable or meaningless when the mean is zero or near zero.

## M04 Standardization, standard errors and coefficient of variation

DESCRIPTIVE STATISTICS · CSSBB · CQE · CMBB · ★ High-yield

$$z = \frac{x - \mu}{\sigma}$$

$$SE(\bar{x}) = \frac{\sigma}{\sqrt{n}} \approx \frac{s}{\sqrt{n}}$$

$$SE(\hat{p}) = \sqrt{\frac{p(1-p)}{n}} \approx \sqrt{\frac{\hat{p}(1-\hat{p})}{n}}$$

$$SE(\bar{x}_1 - \bar{x}_2) = \sqrt{\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}}$$

**WHEN TO USE**

Place observations on a standardized scale and quantify sampling variability.

**EXAM TRAP**

A standard deviation describes individual observations; a standard error describes a statistic across repeated samples.

## M05 Skewness and excess kurtosis

DESCRIPTIVE STATISTICS · CSSBB · CQE

$$g_1 = \frac{n}{(n-1)(n-2)} \sum_{i=1}^{n} \left( \frac{x_i - \bar{x}}{s} \right)^3$$

$$g_2 = \frac{n(n+1)}{(n-1)(n-2)(n-3)} \sum_{i=1}^{n} \left( \frac{x_i - \bar{x}}{s} \right)^4 - \frac{3(n-1)^2}{(n-2)(n-3)}$$

**WHEN TO USE**

Describe distribution shape using bias-adjusted sample estimators.

**EXAM TRAP**

Software packages differ on whether they report kurtosis or excess kurtosis. Normal excess kurtosis is zero.

## M06 Sample-size estimation and finite-population correction

SAMPLING · CSSBB · CQE · CMBB · ★ High-yield

$$n_\mu = \left( \frac{z_{\alpha/2}\sigma}{E} \right)^2$$

$$n_p = \frac{z_{\alpha/2}^2 p(1-p)}{E^2}$$

$$n_0 \ (p \text{ unknown}) = \frac{z_{\alpha/2}^2}{4E^2}$$

$$n_{\text{FPC}} = \frac{n_0}{1 + \dfrac{n_0 - 1}{N}}$$

$$n_h^{\text{proportional}} = n\frac{N_h}{N}$$

$$n_h^{\text{Neyman}} = n\frac{N_h S_h}{\sum_j N_j S_j}$$

**WHEN TO USE**

Plan surveys and measurement studies to achieve a stated margin of error.

**EXAM TRAP**

Round required sample sizes upward. The initial variance or proportion estimate must be defensible.

## M07 Takt time, Little's Law, throughput and process-cycle efficiency

LEAN FLOW · CSSBB · CQE · CMBB · ★ High-yield

$$\text{Takt Time} = \frac{\text{Net Available Production Time}}{\text{Customer Demand}}$$

$$WIP = TH \times CT$$

$$CT = \frac{WIP}{TH}$$

$$TH = \frac{WIP}{CT}$$

$$PCE = \frac{\text{Value-Added Time}}{\text{Total Lead Time}} \times 100\%$$

**WHEN TO USE**

Relate inventory, throughput and lead time and compare production pace with demand.

**EXAM TRAP**

Little's Law requires a stable long-run system and compatible units.

## M08 Yield, defects and rolled throughput yield

SIX SIGMA METRICS · CSSBB · CQE · CMBB · ★ High-yield

$$DPU = \frac{D}{U}$$

$$DPO = \frac{D}{UO}$$

$$DPMO = DPO \times 10^6$$

$$Y = 1 - \frac{\text{Defective Units}}{U}$$

$$Y_{\text{Poisson}} = e^{-DPU}$$

$$RTY = \prod_{i=1}^{k} FPY_i$$

$$DPU = -\ln(RTY)$$

**WHEN TO USE**

Measure defect density, first-pass performance and accumulated hidden-factory losses.

**EXAM TRAP**

A defect is not the same as a defective unit. A unit can contain multiple defects.

## M09 Sigma-level conversion

SIX SIGMA METRICS · CSSBB · CQE · CMBB · ★ High-yield

$$Y = 1 - \frac{DPMO}{10^6}$$

$$Z_{\text{bench}} = \Phi^{-1}(Y)$$

$$Z_{\text{long-term convention}} \approx Z_{\text{bench}} + 1.5$$

**WHEN TO USE**

Convert a one-sided opportunity yield into a benchmark sigma level.

**EXAM TRAP**

The 1.5-sigma shift is a convention, not a universal statistical law. State whether it has been applied.

## M10 Bias, linearity and stability

MSA · CSSBB · CQE · CMBB · ★ High-yield

$$\text{Bias} = \bar{x}_{\text{observed}} - x_{\text{reference}}$$

$$\%\text{Bias} = \frac{\text{Bias}}{\text{Tolerance}} \times 100\%$$

$$\text{Linearity Model} = b_0 + b_1 x_{\text{reference}}$$

$$t_{\text{bias}} = \frac{\bar{d}}{s_d/\sqrt{n}}$$

**WHEN TO USE**

Determine whether a measurement system has systematic error across its operating range.

**EXAM TRAP**

Stability is evaluated over time. Linearity is change in bias across the measurement range.

## M11 Range-method variable gage R&R

MSA | CSSBB | CQE | CMBB | ★ High-yield

$$\hat{\sigma}_e = \frac{\bar{\bar{R}}}{d_2}$$

$$\hat{\sigma}_o = \sqrt{\left(\frac{R_{\bar{x},o}}{d_{2,o}}\right)^2 - \frac{\hat{\sigma}_e^2}{pr}}$$

$$\sigma_{GRR} = \sqrt{\sigma_e^2 + \sigma_o^2}$$

$$\sigma_T = \sqrt{\sigma_{GRR}^2 + \sigma_P^2}$$

**WHEN TO USE**
Estimate repeatability and reproducibility using operator ranges and chart constants.

**EXAM TRAP**
Different AIAG editions use K constants that embed $d_2$ and study-size factors. Do not mix K-factor and $d_2$ forms.

## M12 Crossed ANOVA gage R&R variance components

MSA | CSSBB | CQE | CMBB | ★ High-yield

$$\hat{\sigma}_e^2 = MS_E$$

$$\hat{\sigma}_{PO}^2 = \max\left(0, \frac{MS_{PO} - MS_E}{r}\right)$$

$$\hat{\sigma}_O^2 = \max\left(0, \frac{MS_O - MS_{PO}}{pr}\right)$$

$$\hat{\sigma}_P^2 = \max\left(0, \frac{MS_P - MS_{PO}}{or}\right)$$

$$\sigma_{GRR}^2 = \sigma_e^2 + \sigma_O^2 + \sigma_{PO}^2$$

$$\sigma_T^2 = \sigma_{GRR}^2 + \sigma_P^2$$

**WHEN TO USE**
Separate repeatability, operator, interaction and part-to-part variation in a balanced crossed study.

**EXAM TRAP**
Variance components estimated below zero are normally set to zero, but the negative estimate is a signal to review the model and data.

## M13 PTR, percent GRR, discrimination and ndc

MSA | CSSBB | CQE | CMBB | ★ High-yield

$$PTR_{5.15} = \frac{5.15\,\sigma_{GRR}}{USL - LSL}$$

$$PTR_{6} = \frac{6\,\sigma_{GRR}}{USL - LSL}$$

$$\%GRR_{\text{study}} = 100\frac{\sigma_{GRR}}{\sigma_T}$$

$$\%GRR_{\text{tolerance}} = 100\frac{6\sigma_{GRR}}{USL - LSL}$$

$$ndc = 1.41\frac{\sigma_P}{\sigma_{GRR}}$$

**WHEN TO USE**
Assess whether measurement variation is sufficiently small for the intended application.

**EXAM TRAP**
PTR may use 5.15 or 6 depending on the source or industry. Never compare results without confirming the convention.

## M14 Percent agreement and Cohen/Fleiss kappa

ATTRIBUTE MSA | CSSBB | CQE | CMBB | ★ High-yield

$$P_o = \frac{\text{Number of Agreements}}{\text{Number of Comparisons}}$$

$$\kappa = \frac{P_o - P_e}{1 - P_e}$$

$$p_j = \frac{1}{Nn} \sum_{i=1}^{N} n_{ij}$$

$$P_i = \frac{\sum_{j=1}^{k} n_{ij}^2 - n}{n(n-1)}$$

$$\bar{P} = \frac{1}{N} \sum_i P_i, \qquad P_e = \sum_j p_j^2$$

$$\kappa_{\text{Fleiss}} = \frac{\bar{P} - P_e}{1 - P_e}$$

**WHEN TO USE**

Measure agreement after removing the agreement expected by chance.

**EXAM TRAP**

A high percent agreement can coexist with a low kappa when category prevalence is highly unbalanced.

## M15 Kendall concordance and Kendall tau-b

ATTRIBUTE MSA | CMBB | ★ High-yield

$$\bar{R} = \frac{m(n+1)}{2}$$

$$S = \sum_{i=1}^{n} (R_i - \bar{R})^2$$

$$T_j = \sum_g (t_{jg}^3 - t_{jg})$$

$$W = \frac{12S}{m^2(n^3 - n) - m\sum_j T_j}$$

$$\tau_b = \frac{C - D}{\sqrt{(C + D + T_x)(C + D + T_y)}}$$

**WHEN TO USE**

Evaluate ordinal agreement or association while accounting for rank order and ties.

**EXAM TRAP**

Kappa treats category disagreements nominally unless weighted. Kendall methods preserve ordinal information.

## M16 Six intraclass correlation coefficient forms

ATTRIBUTE MSA | CMBB | ★ High-yield

$$ICC(1,1) = \frac{BMS - WMS}{BMS + (k-1)WMS}$$

$$ICC(1,k) = \frac{BMS - WMS}{BMS}$$

$$ICC(2,1) = \frac{BMS - EMS}{BMS + (k-1)EMS + \frac{k(JMS - EMS)}{n}}$$

$$ICC(2,k) = \frac{BMS - EMS}{BMS + \frac{JMS - EMS}{n}}$$

$$ICC(3,1) = \frac{BMS - EMS}{BMS + (k-1)EMS}$$

$$ICC(3,k) = \frac{BMS - EMS}{BMS}$$

**WHEN TO USE**
Quantify reliability for single versus averaged ratings and random versus fixed judge models.

**EXAM TRAP**
Choose the ICC form from the study design. Reporting an ICC value without its model and single/average designation is incomplete.

## M17 Sensitivity, specificity, predictive values and likelihood ratios

DIAGNOSTIC ACCURACY | CSSBB | CQE | CMBB | ★ High-yield

$$\text{Sensitivity} = \frac{TP}{TP + FN}$$

$$\text{Specificity} = \frac{TN}{TN + FP}$$

$$PPV = \frac{TP}{TP + FP}$$

$$NPV = \frac{TN}{TN + FN}$$

$$\text{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN}$$

$$LR^{+} = \frac{\text{Sensitivity}}{1 - \text{Specificity}}$$

$$LR^{-} = \frac{1 - \text{Sensitivity}}{\text{Specificity}}$$

**WHEN TO USE**
Evaluate inspection, diagnostic and classification systems.

**EXAM TRAP**
Sensitivity and specificity are conditional on true status; PPV and NPV change with prevalence.

## M18 Propagation of error and combined uncertainty

MEASUREMENT UNCERTAINTY | CQE | CMBB | ★ High-yield

$$y = f(x_1, \ldots, x_k)$$

$$u_y^2 \approx \sum_{i=1}^{k} \left(\frac{\partial f}{\partial x_i}\right)^2 u_i^2 + 2 \sum_{i<j} \frac{\partial f}{\partial x_i} \frac{\partial f}{\partial x_j} \operatorname{Cov}(x_i, x_j)$$

$$u_y^2 \text{ (independent)} \approx \sum_{i} \left(\frac{\partial f}{\partial x_i}\right)^2 u_i^2$$

$$U = k\, u_c$$

**WHEN TO USE**
Estimate uncertainty in a calculated output caused by uncertain measured inputs.

**EXAM TRAP**
Root-sum-square combination assumes independence unless covariance terms are included.

## M19 Nested and destructive MSA variance decomposition

MSA CQE CMBB

$$Y_{ijk} = \mu + P_i + S_{j(i)} + O_k + (PO)_{ik} + \varepsilon_{ijk}$$

$$\sigma^2_T = \sigma^2_P + \sigma^2_{S(P)} + \sigma^2_O + \sigma^2_{PO} + \sigma^2_e$$

$$\sigma^2_{MS} = \sigma^2_O + \sigma^2_{PO} + \sigma^2_e$$

**WHEN TO USE**

Analyze measurements when the same physical unit cannot be measured repeatedly.

**EXAM TRAP**

The exact expected-mean-square equations depend on whether operators, samples and parts are crossed or nested.

# A Analyze

DMAIC PHASE 3 · A01–A52

Probability, inference, modeling, capability, reliability and optimization.

52 FORMULA SETS

★ 42 high-yield

## A01 Probability laws and counting rules

PROBABILITY · CSSBB · CQE · CMBB · ★ High-yield

$$P(A^c) = 1 - P(A)$$

$$P(A \cup B) = P(A) + P(B) - P(A \cap B)$$

$$P(A \cap B) = P(A)P(B \mid A)$$

$$P(A \cap B) \text{ (independent)} = P(A)P(B)$$

$${}_nP_r = \frac{n!}{(n-r)!}$$

$$\binom{n}{r} = \frac{n!}{r!(n-r)!}$$

$$\binom{n}{n_1, \ldots, n_k} = \frac{n!}{n_1! \cdots n_k!}$$

**WHEN TO USE**

Construct probabilities for mutually exclusive, overlapping, conditional and independent events.

**EXAM TRAP**

Mutually exclusive events with nonzero probability cannot be independent.

## A02 Total probability and Bayes' theorem

PROBABILITY · CSSBB · CQE · CMBB · ★ High-yield

$$P(B) = \sum_{i=1}^{k} P(B \mid A_i)P(A_i)$$

$$P(A_j \mid B) = \frac{P(B \mid A_j)P(A_j)}{\sum_{i=1}^{k} P(B \mid A_i)P(A_i)}$$

**WHEN TO USE**

Update source or cause probabilities after observing evidence.

**EXAM TRAP**

The posterior denominator must include every mutually exclusive and collectively exhaustive source.

## A03 Expected value, variance, covariance and linear combinations

PROBABILITY · CSSBB · CQE · CMBB · ★ High-yield

$$E(X) = \sum_x x\,p(x) \quad \text{or} \quad \int x f(x)\,dx$$

$$Var(X) = E(X^2) - [E(X)]^2$$

$$Cov(X, Y) = E(XY) - E(X)E(Y)$$

$$E(aX + bY) = aE(X) + bE(Y)$$

$$Var(aX + bY) = a^2 Var(X) + b^2 Var(Y) + 2ab\,Cov(X, Y)$$

**WHEN TO USE**

Calculate long-run averages and variation of combined random quantities.

**EXAM TRAP**

Variances add directly only for independent or uncorrelated variables.

## A04 Bernoulli and binomial distributions

DISCRETE DISTRIBUTIONS · CSSBB · CQE · CMBB · ★ High-yield

$$P(X=x)_{\text{Bernoulli}} = p^x(1-p)^{1-x}, \quad x \in \{0,1\}$$

$$P(X=x)_{\text{Binomial}} = \binom{n}{x} p^x (1-p)^{n-x}$$

$$E(X) = np$$

$$Var(X) = np(1-p)$$

**WHEN TO USE**

Model independent trials with a constant probability and two possible outcomes.

**EXAM TRAP**

A changing probability, dependent trials or more than two outcomes violates the standard binomial assumptions.

## A05 Geometric and negative-binomial distributions

DISCRETE DISTRIBUTIONS · CQE

$$P(X=x)_{\text{Geometric}} = (1-p)^{x-1}p, \quad x = 1,2,\ldots$$

$$E(X) = \frac{1}{p}, \qquad Var(X) = \frac{1-p}{p^2}$$

$$P(X=x)_{\text{NegBin}} = \binom{x-1}{r-1} p^r (1-p)^{x-r}, \quad x = r, r+1, \ldots$$

$$E(X) = \frac{r}{p}, \qquad Var(X) = \frac{r(1-p)}{p^2}$$

**WHEN TO USE**

Model the number of trials required to obtain the first or r-th success.

**EXAM TRAP**

Some software defines the variable as failures before success rather than total trials. Check the convention.

## A06 Hypergeometric and multinomial distributions

DISCRETE DISTRIBUTIONS · CSSBB · CQE · ★ High-yield

$$P(X=x)_{\text{Hypergeom}} = \frac{\binom{K}{x}\binom{N-K}{n-x}}{\binom{N}{n}}$$

$$E(X) = n\frac{K}{N}$$

$$Var(X) = n\frac{K}{N}\left(1-\frac{K}{N}\right)\frac{N-n}{N-1}$$

$$P(X_1 = x_1, \ldots, X_k = x_k) = \frac{n!}{x_1! \cdots x_k!} \prod_{j=1}^{k} p_j^{x_j}$$

**WHEN TO USE**

Hypergeometric for sampling without replacement; multinomial for more than two categories.

**EXAM TRAP**

Using the binomial distribution for a large sampling fraction ignores finite-population dependence.

## A07 Poisson distribution and rate scaling

DISCRETE DISTRIBUTIONS · CSSBB · CQE · CMBB · ★ High-yield

$$P(X=x) = e^{-\lambda}\frac{\lambda^x}{x!}$$

$$E(X) = \lambda$$

$$Var(X) = \lambda$$

$$\lambda_t = rt$$

**WHEN TO USE**

Model independent event counts occurring at an approximately constant rate.

**EXAM TRAP**

The Poisson parameter is the expected count for the exposure interval, not always a per-unit rate.

## A08 Uniform and triangular distributions

CONTINUOUS DISTRIBUTIONS · CQE · CMBB

$$f(x)_{\text{Uniform}} = \frac{1}{b-a}, \quad a \le x \le b$$

$$E(X) = \frac{a+b}{2}$$

$$Var(X) = \frac{(b-a)^2}{12}$$

$$E(X)_{\text{Triangular}} = \frac{a+b+c}{3}$$

$$Var(X)_{\text{Triangular}} = \frac{a^2+b^2+c^2-ab-ac-bc}{18}$$

**WHEN TO USE**

Represent bounded uncertainty when all values are equally likely or when a most-likely value is available.

**EXAM TRAP**

The triangular parameter c is the mode and must satisfy a ≤ c ≤ b.

## A09 Normal and lognormal distributions

CONTINUOUS DISTRIBUTIONS · CSSBB · CQE · CMBB · ★ High-yield

$$f(x)_{\text{Normal}} = \frac{1}{\sigma\sqrt{2\pi}} \exp\left[-\frac{(x-\mu)^2}{2\sigma^2}\right]$$

$$Z = \frac{X-\mu}{\sigma}$$

$$\ln X \sim N(\mu_L, \sigma_L^2)$$

$$E(X)_{\text{Lognormal}} = e^{\mu_L + \sigma_L^2/2}$$

$$Var(X)_{\text{Lognormal}} = \left(e^{\sigma_L^2} - 1\right) e^{2\mu_L + \sigma_L^2}$$

**WHEN TO USE**

Normal for symmetric continuous variation; lognormal for positive right-skewed multiplicative data.

**EXAM TRAP**

The lognormal μ and σ are parameters of ln(X), not the mean and standard deviation of X itself.

## A10 Exponential and Weibull distributions

RELIABILITY DISTRIBUTIONS · CSSBB · CQE · CMBB · ★ High-yield

$$f(t)_{\text{Exponential}} = \lambda e^{-\lambda t}$$

$$R(t) = e^{-\lambda t}$$

$$F(t) = 1 - e^{-\lambda t}$$

$$MTTF = \frac{1}{\lambda}$$

$$F(t)_{\text{Weibull}} = 1 - \exp\left[-\left(\frac{t}{\eta}\right)^{\beta}\right]$$

$$R(t) = \exp\left[-\left(\frac{t}{\eta}\right)^{\beta}\right]$$

$$h(t) = \frac{\beta}{\eta}\left(\frac{t}{\eta}\right)^{\beta-1}$$

$$MTTF = \eta\Gamma\left(1 + \frac{1}{\beta}\right)$$

**WHEN TO USE**

Model time to failure, including decreasing, constant and increasing hazard behavior.

**EXAM TRAP**

β<1 indicates decreasing hazard, β=1 constant hazard and β>1 increasing hazard.

## A11 Gamma and beta distributions

CONTINUOUS DISTRIBUTIONS · CQE · CMBB

$$f(x)_{\text{Gamma}} = \frac{x^{\alpha-1}e^{-x/\theta}}{\Gamma(\alpha)\theta^{\alpha}}, \quad x > 0$$

$$E(X) = \alpha\theta, \qquad Var(X) = \alpha\theta^2$$

$$f(x)_{\text{Beta}} = \frac{x^{\alpha-1}(1-x)^{\beta-1}}{B(\alpha,\beta)}, \quad 0 < x < 1$$

$$E(X) = \frac{\alpha}{\alpha+\beta}$$

$$Var(X) = \frac{\alpha\beta}{(\alpha+\beta)^2(\alpha+\beta+1)}$$

**WHEN TO USE**

Gamma for positive waiting-time variables and beta for bounded proportions or probabilities.

**EXAM TRAP**

Gamma distributions may be parameterized with scale θ or rate λ=1/θ. Confirm which parameterization is being used.

## A12 Student t, chi-square and F statistics

SAMPLING DISTRIBUTIONS · CSSBB · CQE · CMBB · ★ High-yield

$$t = \frac{\bar{X}-\mu}{S/\sqrt{n}}, \qquad df = n-1$$

$$\chi^2 = \frac{(n-1)S^2}{\sigma^2}, \qquad df = n-1$$

$$F = \frac{S_1^2}{S_2^2}, \qquad df_1 = n_1 - 1,\ df_2 = n_2 - 1$$

**WHEN TO USE**

Construct tests and intervals for means and variances under normal-model assumptions.

**EXAM TRAP**

Put the variance specified by the chosen alternative in the numerator of an F statistic so the rejection region is interpreted correctly.

## A13 Central limit theorem and distribution approximations

PROBABILITY · CSSBB · CQE · ★ High-yield

$$\bar{X} \dot{\sim} N\left(\mu, \frac{\sigma^2}{n}\right)$$

$$X_{\text{Binomial}} \dot{\sim} N\left(np, np(1-p)\right)$$

$$X_{\text{Poisson}} \dot{\sim} N(\lambda, \lambda)$$

**WHEN TO USE**

Approximate sampling distributions and large-count discrete distributions.

**EXAM TRAP**

Apply a continuity correction when approximating an integer-valued distribution with a continuous normal distribution.

## A14 Within-process capability indices

PROCESS CAPABILITY · CSSBB · CQE · CMBB · ★ High-yield

$$C_p = \frac{USL - LSL}{6\sigma_w}$$

$$C_{pu} = \frac{USL - \mu}{3\sigma_w}$$

$$C_{pl} = \frac{\mu - LSL}{3\sigma_w}$$

$$C_{pk} = \min(C_{pu}, C_{pl})$$

$$C_{pm} = \frac{USL - LSL}{6\sqrt{\sigma_w^2 + (\mu - T)^2}}$$

**WHEN TO USE**

Evaluate short-term potential and actual capability after establishing statistical control.

**EXAM TRAP**

Cp ignores centering. A process may have an excellent Cp and poor Cpk.

## A15 Overall process-performance indices

PROCESS CAPABILITY | CSSBB | CQE | CMBB | ★ High-yield

$$P_p = \frac{USL - LSL}{6s_{\text{overall}}}$$

$$P_{pu} = \frac{USL - \bar{x}}{3s_{\text{overall}}}$$

$$P_{pl} = \frac{\bar{x} - LSL}{3s_{\text{overall}}}$$

$$P_{pk} = \min(P_{pu}, P_{pl})$$

$$P_{pm} = \frac{USL - LSL}{6\sqrt{s^2_{\text{overall}} + (\bar{x} - T)^2}}$$

**WHEN TO USE**

Summarize long-term performance using total observed variation.

**EXAM TRAP**

Pp/Ppk can be calculated for an unstable process, but instability limits their predictive meaning.

## A16 Normal defect probability and nonnormal percentile capability

PROCESS CAPABILITY | CSSBB | CQE | CMBB | ★ High-yield

$$P(\text{nonconforming}) = \Phi\left(\frac{LSL - \mu}{\sigma}\right) + 1 - \Phi\left(\frac{USL - \mu}{\sigma}\right)$$

$$C_{np} = \frac{USL - LSL}{P_{99.865} - P_{0.135}}$$

$$C_{npu} = \frac{USL - P_{50}}{P_{99.865} - P_{50}}$$

$$C_{npl} = \frac{P_{50} - LSL}{P_{50} - P_{0.135}}$$

$$C_{npk} = \min(C_{npu}, C_{npl})$$

**WHEN TO USE**

Estimate conformance for normal data or assess capability using fitted nonnormal percentiles.

**EXAM TRAP**

Do not apply normal capability equations to strongly nonnormal data without a justified transformation or fitted distribution.

## A17 Confidence interval for Cp

PROCESS CAPABILITY | CQE | CSSBB

$$LCL(C_p) = \hat{C}_p\sqrt{\frac{\chi^2_{\alpha/2,n-1}}{n-1}}$$

$$UCL(C_p) = \hat{C}_p\sqrt{\frac{\chi^2_{1-\alpha/2,n-1}}{n-1}}$$

**WHEN TO USE**

Quantify uncertainty in a normally distributed process capability estimate.

**EXAM TRAP**

This interval assumes independent normal data and a stable estimate of within-process variation.

## A18 Pearson, Spearman and Fisher correlation calculations

CORRELATION CSSBB CQE CMBB ★ High-yield

$$r = \frac{\sum_i (x_i - \bar{x})(y_i - \bar{y})}{\sqrt{\sum_i (x_i - \bar{x})^2 \sum_i (y_i - \bar{y})^2}}$$

$$r_s = 1 - \frac{6 \sum_i d_i^2}{n(n^2 - 1)} \quad \text{(no ties)}$$

$$z_r = \frac{1}{2} \ln\left(\frac{1 + r}{1 - r}\right)$$

$$SE(z_r) = \frac{1}{\sqrt{n - 3}}$$

**WHEN TO USE**

Measure linear or monotonic association and construct an interval for population correlation.

**EXAM TRAP**

Correlation does not establish causation and can be distorted by outliers or restricted ranges.

## A19 Simple linear regression and coefficient of determination

REGRESSION CSSBB CQE CMBB ★ High-yield

$$\hat{y} = b_0 + b_1 x$$

$$b_1 = \frac{\sum_i (x_i - \bar{x})(y_i - \bar{y})}{\sum_i (x_i - \bar{x})^2}$$

$$b_0 = \bar{y} - b_1 \bar{x}$$

$$SST = \sum_i (y_i - \bar{y})^2$$

$$SSE = \sum_i (y_i - \hat{y}_i)^2$$

$$SSR = \sum_i (\hat{y}_i - \bar{y})^2$$

$$R^2 = \frac{SSR}{SST} = 1 - \frac{SSE}{SST}$$

**WHEN TO USE**

Model a continuous response as a linear function of one predictor.

**EXAM TRAP**

A statistically significant slope does not guarantee useful prediction or an adequate residual pattern.

## A20 Regression inference and prediction

REGRESSION CSSBB CQE CMBB ★ High-yield

$$MSE = \frac{SSE}{n - 2}$$

$$s = \sqrt{MSE}$$

$$SE(b_1) = \frac{s}{\sqrt{\sum_i (x_i - \bar{x})^2}}$$

$$t = \frac{b_1 - \beta_{1,0}}{SE(b_1)}$$

$$SE_{\text{mean at } x_0} = s \sqrt{\frac{1}{n} + \frac{(x_0 - \bar{x})^2}{\sum_i (x_i - \bar{x})^2}}$$

$$SE_{\text{prediction at } x_0} = s \sqrt{1 + \frac{1}{n} + \frac{(x_0 - \bar{x})^2}{\sum_i (x_i - \bar{x})^2}}$$

**WHEN TO USE**

Test the slope and distinguish confidence intervals for a mean response from prediction intervals for a new observation.

**EXAM TRAP**

A prediction interval is wider because it includes both mean-estimation uncertainty and individual-process variation.

## A21 Multiple regression in matrix form

ADVANCED REGRESSION | CSSBB | CQE | CMBB | ★ High-yield

$$\mathbf{y} = \mathbf{X}\boldsymbol{\beta} + \boldsymbol{\varepsilon}$$

$$\hat{\boldsymbol{\beta}} = (\mathbf{X}^{\mathsf{T}}\mathbf{X})^{-1}\mathbf{X}^{\mathsf{T}}\mathbf{y}$$

$$\hat{\mathbf{y}} = \mathbf{X}\hat{\boldsymbol{\beta}}$$

$$\mathbf{e} = \mathbf{y} - \hat{\mathbf{y}}$$

$$\widehat{Var}(\hat{\boldsymbol{\beta}}) = MSE(\mathbf{X}^{\mathsf{T}}\mathbf{X})^{-1}$$

$$R^2_{\text{adj}} = 1 - (1 - R^2)\frac{n-1}{n-p}$$

**WHEN TO USE**

Estimate the effects of several predictors on a continuous response.

**EXAM TRAP**

Here p is the number of fitted parameters including the intercept. Some texts instead use k for predictors and write n−k−1.

## A22 VIF, leverage, studentized residual, Cook's distance, PRESS and Mallows Cp

ADVANCED REGRESSION | CSSBB | CQE | CMBB | ★ High-yield

$$VIF_j = \frac{1}{1 - R^2_j}$$

$$h_{ii} = \mathbf{x}_i^{\mathsf{T}}(\mathbf{X}^{\mathsf{T}}\mathbf{X})^{-1}\mathbf{x}_i$$

$$r_i = \frac{e_i}{s\sqrt{1 - h_{ii}}}$$

$$D_i = \frac{e_i^2}{p\,MSE}\,\frac{h_{ii}}{(1 - h_{ii})^2}$$

$$PRESS = \sum_i \left(\frac{e_i}{1 - h_{ii}}\right)^2$$

$$R^2_{\text{pred}} = 1 - \frac{PRESS}{SST}$$

$$C_p^{\text{Mallows}} = \frac{SSE_p}{MSE_{\text{full}}} - n + 2p$$

**WHEN TO USE**

Diagnose multicollinearity, high leverage, influential observations and weak predictive models.

**EXAM TRAP**

Mallows Cp is unrelated to the process-capability index Cp. Interpret it only in the regression-model-selection context.

## A23 Binary, multinomial and ordinal logistic regression

LOGISTIC REGRESSION | CSSBB | CQE | CMBB | ★ High-yield

$$\eta = \beta_0 + \sum_{j=1}^{k} \beta_j x_j$$

$$\log\left(\frac{p}{1-p}\right) = \eta$$

$$p = \frac{e^{\eta}}{1+e^{\eta}} = \frac{1}{1+e^{-\eta}}$$

$$\text{Odds} = \frac{p}{1-p}$$

$$OR_j = e^{\beta_j}$$

$$\log\frac{P(Y=j)}{P(Y=J)} = \alpha_j + \mathbf{x}^{\mathsf{T}}\boldsymbol{\beta}_j$$

$$\log\frac{P(Y\le j)}{P(Y>j)} = \alpha_j - \mathbf{x}^{\mathsf{T}}\boldsymbol{\beta}$$

**WHEN TO USE**

Model binary, nominal or ordered categorical responses.

**EXAM TRAP**

A coefficient changes log-odds. The odds ratio is obtained by exponentiating the coefficient.

## A24 Alternative logistic link functions

LOGISTIC REGRESSION | CMBB

$$g_{\text{logit}}(p) = \ln\frac{p}{1-p}$$

$$g_{\text{probit}}(p) = \Phi^{-1}(p)$$

$$g_{\text{cloglog}}(p) = \ln[-\ln(1-p)]$$

$$g_{\text{log-log}}(p) = -\ln[-\ln(p)]$$

**WHEN TO USE**

Transform a probability on (0,1) to an unbounded model scale.

**EXAM TRAP**

Software naming for gompit, log-log and complementary log-log can differ. Verify the event coding and inverse-link definition.

## A25 Nonlinear least squares and Gauss–Newton update

ADVANCED REGRESSION | CMBB

$$y_i = f(\mathbf{x}_i, \boldsymbol{\beta}) + \varepsilon_i$$

$$SSE(\boldsymbol{\beta}) = \sum_{i=1}^{n}[y_i - f(\mathbf{x}_i, \boldsymbol{\beta})]^2$$

$$\boldsymbol{\beta}^{(r+1)} = \boldsymbol{\beta}^{(r)} + (\mathbf{J}^{\mathsf{T}}\mathbf{J})^{-1}\mathbf{J}^{\mathsf{T}}[\mathbf{y} - \mathbf{f}(\boldsymbol{\beta}^{(r)})]$$

**WHEN TO USE**

Estimate parameters when the response is nonlinear in the model parameters.

**EXAM TRAP**

A model can be curved in x and still be linear in its parameters. Nonlinear regression refers to parameter nonlinearity.

## A26 General linear model, ANOVA and ANCOVA form

GENERAL LINEAR MODEL | CSSBB | CQE | CMBB | ★ High-yield

$$\mathbf{y} = \mathbf{X}\boldsymbol{\beta} + \boldsymbol{\varepsilon}$$

$$\hat{\boldsymbol{\beta}} = (\mathbf{X}^{\mathsf{T}}\mathbf{X})^{-1}\mathbf{X}^{\mathsf{T}}\mathbf{y}$$

$$Y_{ij} = \mu + \tau_i + \varepsilon_{ij} \quad \text{(one-way ANOVA)}$$

$$Y_{ij} = \mu + \tau_i + \gamma(X_{ij} - \bar{X}_{\cdot\cdot}) + \varepsilon_{ij} \quad \text{(ANCOVA)}$$

**WHEN TO USE**

Represent ANOVA, ANCOVA and regression within a common least-squares framework.

**EXAM TRAP**

The CMBB term is general linear model, not generalized linear model. Its response is normally modeled as continuous.

## A27 Nested variance-components model

COMPONENTS OF VARIATION | CSSBB | CQE | CMBB | ★ High-yield

$$Y_{ijk} = \mu + A_i + B_{j(i)} + \varepsilon_{k(ij)}$$

$$\sigma_T^2 = \sigma_A^2 + \sigma_{B(A)}^2 + \sigma_e^2$$

$$\hat{\sigma}_e^2 = MS_E$$

$$\hat{\sigma}_{B(A)}^2 = \frac{MS_{B(A)} - MS_E}{n}$$

$$\hat{\sigma}_A^2 = \frac{MS_A - MS_{B(A)}}{bn}$$

**WHEN TO USE**

Separate hierarchical sources of variation in a balanced nested design.

**EXAM TRAP**

The divisors n and bn depend on the number of lower-level observations in the balanced hierarchy.

## A28 Autocorrelation and Ljung–Box statistic

TIME SERIES | CMBB | ★ High-yield

$$r_k = \frac{\sum_{t=k+1}^{n}(y_t - \bar{y})(y_{t-k} - \bar{y})}{\sum_{t=1}^{n}(y_t - \bar{y})^2}$$

$$Q = n(n+2)\sum_{k=1}^{m}\frac{r_k^2}{n-k}$$

$$\text{Approximate ACF Bounds} \approx \pm\frac{z_{\alpha/2}}{\sqrt{n}}$$

**WHEN TO USE**

Identify serial correlation and test whether a set of autocorrelations differs collectively from zero.

**EXAM TRAP**

Ordinary regression and Shewhart calculations can be misleading when observations are autocorrelated.

## A29 Moving averages and exponential smoothing

TIME SERIES | CSSBB | CQE | CMBB | ★ High-yield

$$MA_t = \frac{1}{m}\sum_{j=0}^{m-1} Y_{t-j}$$

$$WMA_t = \sum_{j=0}^{m-1} w_j Y_{t-j}, \qquad \sum_j w_j = 1$$

$$S_t = \alpha Y_t + (1-\alpha)S_{t-1}, \qquad 0 < \alpha < 1$$

$$\hat{Y}_{t+1} = S_t$$

**WHEN TO USE**

Smooth short-term random variation and create one-step forecasts.

**EXAM TRAP**

A larger α responds faster to changes but produces less smoothing.

## A30 Trend, decomposition and forecast-error metrics

TIME SERIES | CSSBB | CQE | CMBB | ★ High-yield

$$\hat{Y}_t = a + bt$$

$$Y_t = T_t + S_t + C_t + I_t \quad \text{(additive)}$$

$$Y_t = T_t S_t C_t I_t \quad \text{(multiplicative)}$$

$$MAD = \frac{1}{n}\sum_t |Y_t - \hat{Y}_t|$$

$$MSE = \frac{1}{n}\sum_t (Y_t - \hat{Y}_t)^2$$

$$RMSE = \sqrt{MSE}$$

$$MAPE = \frac{100\%}{n}\sum_t \left|\frac{Y_t - \hat{Y}_t}{Y_t}\right|$$

**WHEN TO USE**

Separate trend and seasonality and compare forecast accuracy.

**EXAM TRAP**

MAPE is undefined or unstable when actual values equal or approach zero.

## A31 AR, MA and ARIMA models

TIME SERIES | CMBB | ★ High-yield

$$AR(p): \quad Y_t = c + \sum_{i=1}^{p} \phi_i Y_{t-i} + \varepsilon_t$$

$$MA(q): \quad Y_t = \mu + \varepsilon_t + \sum_{j=1}^{q} \theta_j \varepsilon_{t-j}$$

$$\phi(B)(1-B)^d Y_t = c + \theta(B)\varepsilon_t$$

$$\Delta Y_t = (1-B)Y_t = Y_t - Y_{t-1}$$

**WHEN TO USE**

Model time-series dependence using autoregressive, moving-average and differencing terms.

**EXAM TRAP**

The MA in ARIMA means moving-average error terms, not a rolling arithmetic average of observations.

## A32 One-sample mean tests and confidence intervals

HYPOTHESIS TESTING | CSSBB | CQE | CMBB | ★ High-yield

$$Z = \frac{\bar{X} - \mu_0}{\sigma/\sqrt{n}}$$

$$t = \frac{\bar{X} - \mu_0}{S/\sqrt{n}}, \qquad df = n - 1$$

$$CI_\mu = \bar{X} \pm t_{\alpha/2,n-1}\frac{S}{\sqrt{n}}$$

**WHEN TO USE**

Test or estimate a population mean.

**EXAM TRAP**

Use t rather than z when population sigma is unknown, unless a justified large-sample convention is explicitly stated.

## A33 Independent two-sample means: pooled and Welch

HYPOTHESIS TESTING | CSSBB | CQE | CMBB | ★ High-yield

$$t_{\text{pooled}} = \frac{(\bar{X}_1 - \bar{X}_2) - \Delta_0}{S_p\sqrt{\frac{1}{n_1} + \frac{1}{n_2}}}$$

$$S_p^2 = \frac{(n_1 - 1)S_1^2 + (n_2 - 1)S_2^2}{n_1 + n_2 - 2}$$

$$t_{\text{Welch}} = \frac{(\bar{X}_1 - \bar{X}_2) - \Delta_0}{\sqrt{S_1^2/n_1 + S_2^2/n_2}}$$

$$\nu = \frac{(S_1^2/n_1 + S_2^2/n_2)^2}{\frac{(S_1^2/n_1)^2}{n_1 - 1} + \frac{(S_2^2/n_2)^2}{n_2 - 1}}$$

**WHEN TO USE**

Compare two independent population means.

**EXAM TRAP**

Use pooled variance only when equal population variances are defensible. Welch is the safer default otherwise.

## A34 Paired mean test

HYPOTHESIS TESTING | CSSBB | CQE | CMBB | ★ High-yield

$$d_i = x_{1i} - x_{2i}$$

$$\bar{d} = \frac{1}{n}\sum_i d_i$$

$$t = \frac{\bar{d} - \mu_{d,0}}{s_d/\sqrt{n}}, \qquad df = n - 1$$

$$CI_{\mu_d} = \bar{d} \pm t_{\alpha/2,n-1}\frac{s_d}{\sqrt{n}}$$

**WHEN TO USE**

Analyze before/after or naturally matched observations.

**EXAM TRAP**

The normality assumption applies to the differences, not necessarily to each original sample.

## A35 One- and two-proportion inference

HYPOTHESIS TESTING | CSSBB | CQE | CMBB | ★ High-yield

$$z_{\text{one proportion}} = \frac{\hat{p} - p_0}{\sqrt{p_0(1 - p_0)/n}}$$

$$\hat{p}_c = \frac{x_1 + x_2}{n_1 + n_2}$$

$$z_{\text{two-proportion test}} = \frac{(\hat{p}_1 - \hat{p}_2) - \Delta_0}{\sqrt{\hat{p}_c(1 - \hat{p}_c)\left(\frac{1}{n_1} + \frac{1}{n_2}\right)}}$$

$$SE_{\text{two-proportion CI}} = \sqrt{\frac{\hat{p}_1(1 - \hat{p}_1)}{n_1} + \frac{\hat{p}_2(1 - \hat{p}_2)}{n_2}}$$

**WHEN TO USE**

Compare defect rates or other binary-event proportions.

**EXAM TRAP**

Pool the proportions for the standard null test of equality, but not for the usual confidence interval.

## A36 Variance and standard-deviation inference

HYPOTHESIS TESTING | CSSBB | CQE | CMBB | ★ High-yield

$$\chi^2 = \frac{(n-1)S^2}{\sigma_0^2}$$

$$CI_{\sigma^2} = \left[\frac{(n-1)S^2}{\chi^2_{1-\alpha/2,n-1}}, \frac{(n-1)S^2}{\chi^2_{\alpha/2,n-1}}\right]$$

$$F = \frac{S_1^2}{S_2^2}$$

**WHEN TO USE**

Test a process variance or compare two normally distributed process variances.

**EXAM TRAP**

Classical chi-square and F variance procedures are highly sensitive to nonnormality.

## A37 One-way ANOVA

ANOVA | CSSBB | CQE | CMBB | ★ High-yield

$$SS_T = \sum_i \sum_j (y_{ij} - \bar{y}_{..})^2$$

$$SS_B = \sum_i n_i(\bar{y}_{i.} - \bar{y}_{..})^2$$

$$SS_E = \sum_i \sum_j (y_{ij} - \bar{y}_{i.})^2$$

$$SS_T = SS_B + SS_E$$

$$F = \frac{MS_B}{MS_E}$$

**WHEN TO USE**

Test equality of three or more population means.

**EXAM TRAP**

A significant overall F test indicates at least one mean differs; it does not identify which means differ.

## A38 Chi-square goodness-of-fit and independence

CATEGORICAL ANALYSIS | CSSBB | CQE | CMBB | ★ High-yield

$$\chi^2 = \sum_i \frac{(O_i - E_i)^2}{E_i}$$

$$E_{ij} = \frac{(\text{Row Total}_i)(\text{Column Total}_j)}{\text{Grand Total}}$$

$$df_{\text{independence}} = (r-1)(c-1)$$

$$df_{\text{GOF}} = k - 1 - m$$

**WHEN TO USE**

Compare observed categorical counts with expected counts.

**EXAM TRAP**

The degrees of freedom for goodness-of-fit are reduced by the number m of distribution parameters estimated from the data.

## A39 Sign, Mann–Whitney, Wilcoxon and Kruskal–Wallis statistics

NONPARAMETRIC TESTS | CSSBB | CQE

$$P(X = x)_{\text{Sign Test}} = \binom{n}{x}(0.5)^n$$

$$U_1 = n_1 n_2 + \frac{n_1(n_1+1)}{2} - R_1$$

$$U_2 = n_1 n_2 - U_1$$

$$W = \sum \text{signed ranks}$$

$$H = \frac{12}{N(N+1)} \sum_{i=1}^{k} \frac{R_i^2}{n_i} - 3(N+1)$$

**WHEN TO USE**

Compare medians, locations or distributions when parametric assumptions are unsuitable.

**EXAM TRAP**

Many nonparametric tests still assume independent observations and similarly shaped distributions for a pure location interpretation.

## A40 TOST equivalence and approximate sample size for two means

EQUIVALENCE AND POWER · CSSBB · CQE

$$H_{01}: \Delta \le -\delta, \qquad H_{02}: \Delta \ge \delta$$

$$t_L = \frac{\hat{\Delta} + \delta}{SE(\hat{\Delta})}$$

$$t_U = \frac{\hat{\Delta} - \delta}{SE(\hat{\Delta})}$$

$$n_{\text{per group}} \approx \frac{2\sigma^2 (z_{\alpha/2} + z_{\beta})^2}{\Delta^2}$$

**WHEN TO USE**

Demonstrate practical equivalence and plan a two-sample comparison for a desired power.

**EXAM TRAP**

Failing to detect a difference is not evidence of equivalence. Equivalence requires predefined practical limits.

## A41 Probability of acceptance for a single-sampling plan

ACCEPTANCE SAMPLING · CQE · CSSBB · ★ High-yield

$$P_a(p)_{\text{Binomial}} = \sum_{x=0}^{c} \binom{n}{x} p^x (1-p)^{n-x}$$

$$P_a(D)_{\text{Hypergeom}} = \sum_{x=0}^{c} \frac{\binom{D}{x}\binom{N-D}{n-x}}{\binom{N}{n}}$$

$$P_a(p)_{\text{Poisson}} \approx \sum_{x=0}^{c} e^{-np} \frac{(np)^x}{x!}$$

**WHEN TO USE**

Construct an operating-characteristic curve for an attribute sampling plan.

**EXAM TRAP**

Use hypergeometric calculations when sampling without replacement from a finite lot and the sampling fraction is material.

## A42 AOQ, AOQL, ATI and ASN

ACCEPTANCE SAMPLING · CQE · ★ High-yield

$$AOQ(p) = pP_a(p)\frac{N-n}{N}$$

$$AOQL = \max_p AOQ(p)$$

$$ATI(p) = nP_a(p) + N[1 - P_a(p)]$$

$$ASN_{\text{double}} = n_1 + n_2 P(c_1 < X_1 < r_1)$$

**WHEN TO USE**

Evaluate long-run outgoing quality and inspection effort for rectifying inspection plans.

**EXAM TRAP**

AOQ assumes rejected lots are screened and nonconforming units are replaced or corrected.

## A43 Variables sampling standardized distance

ACCEPTANCE SAMPLING · CQE

$$Q_U = \frac{USL - \bar{X}}{S}$$

$$Q_L = \frac{\bar{X} - LSL}{S}$$

$$\text{Accept Upper-Spec Plan if} \quad \bar{X} + kS \le USL$$

$$\text{Accept Lower-Spec Plan if} \quad \bar{X} - kS \ge LSL$$

**WHEN TO USE**

Use continuous measurements rather than defect counts to make lot-acceptance decisions.

**EXAM TRAP**

The acceptance constant k depends on AQL, LTPD, producer risk, consumer risk and whether sigma is known.

## A44 Reliability, failure distribution, hazard and mean life

RELIABILITY · CQE · CSSBB · CMBB · ★ High-yield

$$R(t) = P(T > t) = 1 - F(t)$$

$$f(t) = \frac{dF(t)}{dt} = -\frac{dR(t)}{dt}$$

$$h(t) = \frac{f(t)}{R(t)}$$

$$R(t) = \exp\left[-\int_0^t h(u)\,du\right]$$

$$MTTF = \int_0^\infty R(t)\,dt$$

**WHEN TO USE**

Relate life distribution, survival probability and instantaneous failure rate.

**EXAM TRAP**

Failure probability F(t) and failure density f(t) have different units and meanings.

## A45 Series, parallel, k-out-of-n and standby reliability

RELIABILITY · CQE · CMBB · ★ High-yield

$$R_{\text{series}} = \prod_{i=1}^{n} R_i$$

$$R_{\text{parallel}} = 1 - \prod_{i=1}^{n}(1 - R_i)$$

$$R_{k\text{-of-}n} = \sum_{j=k}^{n} \binom{n}{j} R^j (1-R)^{n-j}$$

$$R_{\text{standby}}(t) = e^{-\lambda t} \sum_{i=0}^{n} \frac{(\lambda t)^i}{i!}$$

**WHEN TO USE**

Calculate system reliability under independent component configurations.

**EXAM TRAP**

The simple standby equation assumes identical exponential components, perfect switching and no failures while in standby.

## A46 Availability and maintainability

RELIABILITY · CQE · CMBB · ★ High-yield

$$A_i = \frac{MTBF}{MTBF + MTTR}$$

$$A_o = \frac{\text{Uptime}}{\text{Uptime} + \text{Downtime}}$$

$$M(t) = 1 - e^{-\mu t}$$

$$MTTR = \frac{1}{\mu}$$

**WHEN TO USE**

Quantify the probability that a repairable system is available and the probability it can be restored by time t.

**EXAM TRAP**

Inherent availability excludes logistics and administrative delay; operational availability includes actual downtime sources.

## A47 Zero-failure reliability demonstration

RELIABILITY · CQE · CMBB

$$n \geq \frac{\ln(1-C)}{\ln(R_0)}$$

$$T_{\text{test}} \geq -\frac{\ln(1-C)}{\lambda_0}$$

$$T_{\text{test}} \geq -MTBF_0 \ln(1-C)$$

**WHEN TO USE**

Plan a no-failure demonstration test for a required reliability or MTBF at confidence C.

**EXAM TRAP**

The exponential-time form assumes a constant failure rate and normally uses total accumulated test time.

## A48 Duane reliability-growth model

RELIABILITY GROWTH | CMBB | ★ High-yield

$$MTBF_c(T) = KT^{\alpha}$$

$$\lambda_c(T) = \frac{1}{KT^{\alpha}}$$

$$\lambda_i(T) = \frac{1-\alpha}{K}T^{-\alpha}$$

$$MTBF_i(T) = \frac{KT^{\alpha}}{1-\alpha} = \frac{MTBF_c(T)}{1-\alpha}$$

$$T = \left[\frac{(1-\alpha)MTBF_i}{K}\right]^{1/\alpha}$$

**WHEN TO USE**

Model reliability improvement as corrective design changes are incorporated during testing.

**EXAM TRAP**

The model assumes corrective changes are incorporated promptly after failures. Delayed fixes violate the interpretation.

## A49 AMSAA/Crow reliability-growth model

RELIABILITY GROWTH | CMBB | ★ High-yield

$$m(T) = \lambda T^{\beta}$$

$$\lambda_c(T) = \frac{m(T)}{T} = \lambda T^{\beta-1}$$

$$\lambda_i(T) = \frac{dm(T)}{dT} = \lambda\beta T^{\beta-1}$$

$$MTBF_i(T) = \frac{1}{\lambda\beta T^{\beta-1}}$$

**WHEN TO USE**

Model cumulative failures and changing failure intensity using a power-law process.

**EXAM TRAP**

β<1 indicates reliability growth, β=1 constant failure intensity and β>1 deterioration.

## A50 Monte Carlo estimate, simulation variance and confidence interval

SIMULATION | CSSBB | CQE | CMBB | ★ High-yield

$$\hat{\mu} = \frac{1}{N}\sum_{i=1}^{N} Y_i$$

$$s_Y^2 = \frac{\sum_i (Y_i - \hat{\mu})^2}{N-1}$$

$$SE(\hat{\mu}) = \frac{s_Y}{\sqrt{N}}$$

$$CI_{\mu} = \hat{\mu} \pm t_{\alpha/2,N-1}\frac{s_Y}{\sqrt{N}}$$

**WHEN TO USE**

Estimate the expected performance and uncertainty of a simulated system.

**EXAM TRAP**

Increasing simulation replications reduces Monte Carlo error but does not correct a structurally invalid model.

## A51 M/M/1 queue and Little's Law

QUEUEING CMBB ★ High-yield

$$\rho = \frac{\lambda}{\mu} < 1$$

$$P_0 = 1 - \rho$$

$$P_n = (1-\rho)\rho^n$$

$$L = \frac{\rho}{1-\rho}$$

$$L_q = \frac{\rho^2}{1-\rho}$$

$$W = \frac{1}{\mu - \lambda}$$

$$W_q = \frac{\rho}{\mu - \lambda}$$

$$L = \lambda W, \qquad L_q = \lambda W_q$$

**WHEN TO USE**

Estimate steady-state congestion for one exponential server and Poisson arrivals.

**EXAM TRAP**

The steady-state formulas require λ<μ. Utilization near one causes waiting time to increase nonlinearly.

## A52 Linear-programming primal and dual forms

OPTIMIZATION CSSBB CQE CMBB ★ High-yield

$$\max_{\mathbf{x}} \quad Z = \mathbf{c}^{\mathsf{T}}\mathbf{x}$$

$$\text{subject to} \quad \mathbf{A}\mathbf{x} \le \mathbf{b}, \qquad \mathbf{x} \ge 0$$

$$\min_{\mathbf{y}} \quad W = \mathbf{b}^{\mathsf{T}}\mathbf{y}$$

$$\text{subject to} \quad \mathbf{A}^{\mathsf{T}}\mathbf{y} \ge \mathbf{c}, \qquad \mathbf{y} \ge 0$$

**WHEN TO USE**

Optimize allocation, scheduling, routing, assignment, capacity and mixture decisions subject to linear constraints.

**EXAM TRAP**

A feasible solution satisfies all constraints; an optimal solution is the best feasible solution. Feasibility does not imply optimality.

# I

DMAIC PHASE 4 · I01–I16

# Improve

DOE, robust design, Lean calculations and solution optimization.

16 FORMULA SETS

★ 11 high-yield

## I01 Two-level factorial coded model and main effect

DOE CSSBB CQE CMBB ★ High-yield

$$x_i = \frac{X_i - X_{0i}}{\Delta X_i}$$

$$\text{Effect of } A = \bar{y}_{A+} - \bar{y}_{A-}$$

$$\hat{y} = b_0 + \sum_i b_i x_i + \sum_{i<j} b_{ij} x_i x_j + \cdots$$

$$b_A = \frac{\text{Effect of } A}{2}$$

**WHEN TO USE**

Express factorial experiments in coded units and estimate factor effects.

**EXAM TRAP**

In a ±1 coded model, the regression coefficient is half the corresponding effect.

## I02 Factorial contrasts and sums of squares

DOE CSSBB CQE CMBB ★ High-yield

$$C_E = \sum_i s_i T_i$$

$$\text{Effect} = \frac{C_E}{r 2^{k-1}}$$

$$SS_E = \frac{C_E^2}{r 2^k}$$

$$MS_E = \frac{SS_E}{df_E}$$

$$F_E = \frac{MS_E}{MS_{\text{error}}}$$

**WHEN TO USE**

Calculate effects and ANOVA components for a balanced replicated $2^k$ factorial design.

**EXAM TRAP**

The contrast divisor changes with replication and the number of factors.

## I03 Randomized complete block and Latin-square models

DOE CSSBB CQE CMBB ★ High-yield

$$Y_{ij} = \mu + \tau_i + \beta_j + \varepsilon_{ij} \quad \text{(RCBD)}$$

$$Y_{ijk} = \mu + \rho_i + \kappa_j + \tau_k + \varepsilon_{ijk} \quad \text{(Latin square)}$$

$$SS_{\text{Treatment}} = \sum_i \frac{T_i^2}{b} - \frac{G^2}{N}$$

**WHEN TO USE**

Remove nuisance variation through blocking while comparing treatments.

**EXAM TRAP**

Blocking improves precision only when the block variable explains meaningful variation.

## I04 Balanced incomplete block design identities

ADVANCED DOE CMBB ★ High-yield

$$bk = vr$$

$$r(k-1) = \lambda(v-1)$$

$$\lambda = \frac{r(k-1)}{v-1}$$

$$N = bk = vr$$

**WHEN TO USE**

Verify whether an incomplete-block structure is balanced.

**EXAM TRAP**

v is the number of treatments, b blocks, k treatments per block, r replications per treatment and λ joint appearances per treatment pair.

## I05 Second-order response-surface model and stationary point

RESPONSE SURFACE | CSSBB | CQE | CMBB | ★ High-yield

$$\hat{y} = \beta_0 + \sum_{i=1}^{k} \beta_i x_i + \sum_{i=1}^{k} \beta_{ii} x_i^2 + \sum_{i<j} \beta_{ij} x_i x_j$$

$$\hat{y} = \beta_0 + \mathbf{b}^{\mathsf{T}}\mathbf{x} + \mathbf{x}^{\mathsf{T}}\mathbf{B}\mathbf{x}$$

$$\mathbf{x}_s = -\frac{1}{2}\mathbf{B}^{-1}\mathbf{b}$$

**WHEN TO USE**
Model curvature and locate a candidate optimum or saddle point.

**EXAM TRAP**
A stationary point may be a maximum, minimum or saddle. Examine eigenvalues or canonical form.

## I06 Mixture constraint and Scheffé models

MIXTURE DOE | CMBB | ★ High-yield

$$\sum_{i=1}^{q} x_i = 1, \qquad x_i \geq 0$$

$$E(Y)_{\text{linear}} = \sum_{i} \beta_i x_i$$

$$E(Y)_{\text{quadratic}} = \sum_{i} \beta_i x_i + \sum_{i<j} \beta_{ij} x_i x_j$$

$$E(Y)_{\text{special cubic}} = \sum_{i} \beta_i x_i + \sum_{i<j} \beta_{ij} x_i x_j + \sum_{i<j<k} \beta_{ijk} x_i x_j x_k$$

**WHEN TO USE**
Model formulations whose component proportions must sum to a constant.

**EXAM TRAP**
A standard factorial model is inappropriate because mixture components cannot vary independently.

## I07 D-optimal design criterion

OPTIMAL DESIGN | CMBB

$$D\text{-optimality:} \qquad \max \left|\mathbf{X}^{\mathsf{T}}\mathbf{X}\right|$$

$$Var(\hat{\beta}) = \sigma^2(\mathbf{X}^{\mathsf{T}}\mathbf{X})^{-1}$$

**WHEN TO USE**
Select experimental runs that minimize the generalized variance of coefficient estimates.

**EXAM TRAP**
A D-optimal design is model-dependent. Changing the assumed model can change the selected runs.

## I08 Split-plot model and two error strata

SPLIT-PLOT DOE | CMBB

$$Y_{ijk} = \mu + A_i + W_{k(i)} + B_j + (AB)_{ij} + \varepsilon_{ijk}$$

$$Var(W_{k(i)}) = \sigma_W^2$$

$$Var(\varepsilon_{ijk}) = \sigma_e^2$$

**WHEN TO USE**
Analyze experiments containing hard-to-change whole-plot factors and easy-to-change subplot factors.

**EXAM TRAP**
Whole-plot effects and subplot effects use different error terms. Treating all runs as completely randomized gives invalid tests.

## I09 Taguchi signal-to-noise ratios

ROBUST DESIGN · CSSBB · CQE · CMBB · ★ High-yield

$$S/N_{\text{smaller}} = -10\log_{10}\left(\frac{1}{n}\sum_i y_i^2\right)$$

$$S/N_{\text{larger}} = -10\log_{10}\left(\frac{1}{n}\sum_i \frac{1}{y_i^2}\right)$$

$$S/N_{\text{nominal}} = 10\log_{10}\left(\frac{\bar{y}^2}{s^2}\right)$$

**WHEN TO USE**
Select factor settings that reduce sensitivity to noise.

**EXAM TRAP**
The preferred setting maximizes S/N regardless of which S/N definition is used.

## I10 Local gradient and steepest-ascent step

OPTIMIZATION · CSSBB · CMBB

$$\nabla\hat{y} = \begin{bmatrix} \partial\hat{y}/\partial x_1 \\ \vdots \\ \partial\hat{y}/\partial x_k \end{bmatrix}$$

$$\Delta\mathbf{x} = c\,\nabla\hat{y} \quad \text{(ascent)}$$

$$\Delta\mathbf{x} = -c\,\nabla\hat{y} \quad \text{(descent)}$$

**WHEN TO USE**
Move from a first-order experimental region toward improved response values.

**EXAM TRAP**
The linear gradient is only locally valid. Stop and refit when curvature becomes evident.

## I11 OEE and the six big losses framework

LEAN · CSSBB · CQE · CMBB · ★ High-yield

$$Availability = \frac{\text{Run Time}}{\text{Planned Production Time}}$$

$$Performance = \frac{\text{Ideal Cycle Time} \times \text{Total Count}}{\text{Run Time}}$$

$$Quality = \frac{\text{Good Count}}{\text{Total Count}}$$

$$OEE = Availability \times Performance \times Quality$$

**WHEN TO USE**
Separate equipment loss into availability, speed and quality components.

**EXAM TRAP**
Use proportions, not percentages entered as whole numbers, when multiplying the three OEE components.

## I12 Kanban sizing and line balancing

LEAN · CSSBB · CQE · CMBB · ★ High-yield

$$N_{\text{kanban}} = \frac{D\,L(1+S)}{C}$$

$$N_{\min} = \left\lceil \frac{\sum_i t_i}{CT} \right\rceil$$

$$\text{Line Efficiency} = \frac{\sum_i t_i}{N_{\text{stations}} \times CT} \times 100\%$$

$$\text{Balance Delay} = 1 - \text{Line Efficiency}$$

**WHEN TO USE**
Set pull-system container counts and assess workstation balance.

**EXAM TRAP**
Round Kanban cards upward. Demand and lead time must use compatible units.

## I13 Throughput-accounting measures

THEORY OF CONSTRAINTS · CSSBB · CMBB

$$T = \text{Sales} - \text{Totally Variable Cost}$$

$$NP = T - OE$$

$$ROI = \frac{T - OE}{I}$$

$$\text{Productivity} = \frac{T}{OE}$$

$$\text{Investment Turns} = \frac{T}{I}$$

**WHEN TO USE**

Evaluate system improvement from the constraint-focused throughput perspective.

**EXAM TRAP**

Throughput accounting usually treats direct material as the primary totally variable cost; do not substitute conventional absorption-cost assumptions automatically.

## I14 Worst-case and RSS tolerance stack-up

TOLERANCE DESIGN · CQE · CSSBB · CMBB · ★ High-yield

$$T_{\text{worst case}} = \sum_{i=1}^{k} |a_i| T_i$$

$$T_{\text{RSS}} = \sqrt{\sum_{i=1}^{k} (a_i T_i)^2}$$

$$\sigma_Y^2 = \sum_i a_i^2 \sigma_i^2 + 2 \sum_{i<j} a_i a_j Cov(X_i, X_j)$$

**WHEN TO USE**

Estimate assembly tolerance from contributing dimensions.

**EXAM TRAP**

RSS assumes independent, centered and statistically behaving component dimensions unless covariance and mean shifts are explicitly modeled.

## I15 Taguchi quality-loss function

ROBUST DESIGN · CSSBB · CQE · CMBB · ★ High-yield

$$L(y) = k(y - T)^2$$

$$k = \frac{A_0}{\Delta_0^2}$$

$$E[L(Y)] = k\left[\sigma^2 + (\mu - T)^2\right]$$

**WHEN TO USE**

Quantify economic loss caused by both variation and off-target performance.

**EXAM TRAP**

Meeting specification does not imply zero loss. Loss is minimized at the target.

## I16 Diametrical true-position error

GD&T · CQE

$$TP = 2\sqrt{(\Delta x)^2 + (\Delta y)^2}$$

$$\text{Bonus Tolerance} = \text{Actual Size} - MMC \quad \text{(internal feature)}$$

$$\text{Total Position Tolerance} = \text{Specified Position Tolerance} + \text{Bonus Tolerance}$$

**WHEN TO USE**

Evaluate the radial location error of a feature under a diametrical position tolerance.

**EXAM TRAP**

The sign and bonus-tolerance calculation differ for internal and external features.

C

DMAIC PHASE 5 · C01–C11

# Control

SPC, monitoring, feedback control and sustained performance.

11 FORMULA SETS

★ 8 high-yield

## C01 General three-sigma control-limit structure

SPC | CSSBB | CQE | CMBB | ★ High-yield

$$UCL = \mu_{\text{statistic}} + 3\sigma_{\text{statistic}}$$
$$CL = \mu_{\text{statistic}}$$
$$LCL = \mu_{\text{statistic}} - 3\sigma_{\text{statistic}}$$

**WHEN TO USE**

Construct natural process limits for a monitored statistic.

**EXAM TRAP**

Control limits describe process behavior. Specification limits describe customer or engineering requirements.

## C02 X̄–R control charts

VARIABLES CHARTS | CSSBB | CQE | CMBB | ★ High-yield

$$UCL_{\bar{X}} = \bar{\bar{X}} + A_2\bar{R}$$
$$CL_{\bar{X}} = \bar{\bar{X}}$$
$$LCL_{\bar{X}} = \bar{\bar{X}} - A_2\bar{R}$$
$$UCL_R = D_4\bar{R}$$
$$CL_R = \bar{R}$$
$$LCL_R = D_3\bar{R}$$
$$\hat{\sigma}_w = \frac{\bar{R}}{d_2}$$

**WHEN TO USE**

Monitor subgroup means and ranges for relatively small rational subgroup sizes.

**EXAM TRAP**

Assess the R chart before interpreting the X̄ chart because the mean-chart limits depend on a stable variation estimate.

## C03 X̄–S control charts

VARIABLES CHARTS | CSSBB | CQE | CMBB | ★ High-yield

$$UCL_{\bar{X}} = \bar{\bar{X}} + A_3\bar{S}$$
$$LCL_{\bar{X}} = \bar{\bar{X}} - A_3\bar{S}$$
$$UCL_S = B_4\bar{S}$$
$$CL_S = \bar{S}$$
$$LCL_S = B_3\bar{S}$$
$$\hat{\sigma}_w = \frac{\bar{S}}{c_4}$$

**WHEN TO USE**

Monitor subgroup means and standard deviations, particularly for larger subgroup sizes.

**EXAM TRAP**

Use control-chart constants corresponding to the actual subgroup size.

## C04 Individuals and moving-range charts

VARIABLES CHARTS | CSSBB | CQE | CMBB | ★ High-yield

$$MR_i = |X_i - X_{i-1}|$$

$$\hat{\sigma} = \frac{\overline{MR}}{d_2}$$

$$UCL_X = \bar{X} + 3\frac{\overline{MR}}{d_2}$$

$$LCL_X = \bar{X} - 3\frac{\overline{MR}}{d_2}$$

$$UCL_{MR} = D_4\overline{MR}$$

$$LCL_{MR} = D_3\overline{MR}$$

**WHEN TO USE**

Monitor one observation per time period.

**EXAM TRAP**

Successive moving ranges overlap and are not independent; use the established chart method rather than treating them as ordinary subgroups.

## C05 p and np charts

ATTRIBUTE CHARTS | CSSBB | CQE | CMBB | ★ High-yield

$$\bar{p} = \frac{\sum_i d_i}{\sum_i n_i}$$

$$UCL_{p,i} = \bar{p} + 3\sqrt{\frac{\bar{p}(1-\bar{p})}{n_i}}$$

$$LCL_{p,i} = \bar{p} - 3\sqrt{\frac{\bar{p}(1-\bar{p})}{n_i}}$$

$$CL_{np} = n\bar{p}$$

$$UCL_{np} = n\bar{p} + 3\sqrt{n\bar{p}(1-\bar{p})}$$

$$LCL_{np} = n\bar{p} - 3\sqrt{n\bar{p}(1-\bar{p})}$$

**WHEN TO USE**

Monitor the fraction or number of defective units.

**EXAM TRAP**

An np chart requires constant sample size. A p chart can accommodate varying n with varying limits.

## C06 c and u charts

ATTRIBUTE CHARTS | CSSBB | CQE | CMBB | ★ High-yield

$$\bar{c} = \frac{\sum_i c_i}{m}$$

$$UCL_c = \bar{c} + 3\sqrt{\bar{c}}$$

$$LCL_c = \bar{c} - 3\sqrt{\bar{c}}$$

$$\bar{u} = \frac{\sum_i c_i}{\sum_i n_i}$$

$$UCL_{u,i} = \bar{u} + 3\sqrt{\frac{\bar{u}}{n_i}}$$

$$LCL_{u,i} = \bar{u} - 3\sqrt{\frac{\bar{u}}{n_i}}$$

**WHEN TO USE**

Monitor defect counts or defects per inspection unit.

**EXAM TRAP**

Use a c chart only when the area of opportunity is constant. Negative calculated lower limits are set to zero.

## C07 Standardized short-run control chart

SHORT-RUN SPC | CSSBB | CQE | CMBB

$$Z_i = \frac{X_i - T_i}{\sigma_i}$$

$$CL_Z = 0$$

$$UCL_Z = +3$$

$$LCL_Z = -3$$

**WHEN TO USE**

Combine several low-volume products by standardizing each observation to its own target and variation.

**EXAM TRAP**

Targets and sigma estimates must be valid for each product before combining standardized observations.

## C08 Moving-average and EWMA charts

ADVANCED SPC | CSSBB | CQE | CMBB | ★ High-yield

$$MA_i = \frac{1}{w}\sum_{j=0}^{w-1} X_{i-j}$$

$$\sigma_{MA} = \frac{\sigma}{\sqrt{w}}$$

$$Z_i = \lambda X_i + (1-\lambda)Z_{i-1}$$

$$UCL_i = \mu_0 + L\sigma\sqrt{\frac{\lambda}{2-\lambda}[1-(1-\lambda)^{2i}]}$$

$$LCL_i = \mu_0 - L\sigma\sqrt{\frac{\lambda}{2-\lambda}[1-(1-\lambda)^{2i}]}$$

**WHEN TO USE**

Detect smaller sustained shifts by incorporating information from several observations.

**EXAM TRAP**

A small EWMA λ gives more memory and slower response to large abrupt shifts.

## C09 Tabular CUSUM

ADVANCED SPC | CSSBB | CQE | CMBB | ★ High-yield

$$C_i^+ = \max\left[0,\ C_{i-1}^+ + X_i - (\mu_0 + K)\right]$$

$$C_i^- = \max\left[0,\ C_{i-1}^- + (\mu_0 - K) - X_i\right]$$

$$\text{Signal if} \quad C_i^+ > H \quad \text{or} \quad C_i^- > H$$

**WHEN TO USE**

Accumulate small deviations from target to identify sustained process shifts.

**EXAM TRAP**

K is the reference value and H is the decision interval. They are not specification limits.

## C10 Average run length approximations

SPC PERFORMANCE | CSSBB | CQE | CMBB

$$ARL_0 \approx \frac{1}{\alpha}$$

$$ARL_1 \approx \frac{1}{1-\beta}$$

$$ATS = ARL \times \text{Sampling Interval}$$

**WHEN TO USE**

Express the expected number of plotted statistics or time until a signal.

**EXAM TRAP**

These simple formulas apply when each plotting decision has a constant independent signal probability.

## C11 Feedback error, PID controller and closed-loop response

AUTOMATED PROCESS CONTROL CMBB

$$e(t) = r(t) - y(t)$$

$$u(t) = K_c \left[ e(t) + \frac{1}{\tau_I} \int_0^t e(\xi) d\xi + \tau_D \frac{de(t)}{dt} \right]$$

$$G_{CL}(s) = \frac{G_c(s) G_p(s)}{1 + G_c(s) G_p(s)}$$

**WHEN TO USE**

Describe how an automated controller manipulates an input in response to set-point error.

**EXAM TRAP**

APC actively adjusts the process. SPC detects statistical signals; the two methods can be complementary rather than substitutes.